Arash Jalil Khabbazi

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EDUCATION

Purdue University — *PhD, Mechanical Engineering* | GPA 3.9/4.0 West Lafayette, IN | 08/2023 – 08/2028
Concentration in Computational Science and Engineering

- *Research interests:* predictive control, reinforcement learning (RL), smart buildings, energy systems
- *Coursework:* Reinforcement Learning & Control, Applied Optimal Control & Estimation, Convex Optimization, Artificial Intelligence, Statistical Methods, Industrial Internet of Things, Distributed Energy Resources, Analysis of Thermal Systems

University of British Columbia (UBC) — *MS, Mechanical Engineering* | GPA 4.0/4.0 BC, Canada | 05/2021 – 08/2023

University of Tabriz — *BS, Mechanical Engineering* | GPA 4.0/4.0 | Ranked 1 of 155+ EA, Iran | 08/2016 – 08/2020

EXPERIENCE

Purdue University — *Graduate Research Assistant, Kircher Research Group* West Lafayette, IN | 08/2023 – Present

- Identified a grey-box thermal model of a 13-zone campus building from BAS data for predictive pre-cooling control.
- Developed a two-parameter, tuning-free HVAC controller with **up to 87% of optimal savings** in simulation (CDC'26).
- Built data-driven predictive control with the **lowest prediction error and failure rate** of all predictors (under review).
- Derived and tested a multi-zone model showing small HVAC demos **overstate savings** (ACC'26).
- Developed current-limiting control for home appliances to **avoid $2k–$10k panel upgrades** (CISBAT'25, ACEEE'26).
- Led a 4-institution review of **all known** real-world demonstrations of MPC and RL for HVAC (Applied Energy).

Tesla, Inc. — *Applied ML Intern* Sparks, NV | 05/2025 – 08/2025

- Developed MySQL data pipelines and engineered high-signal features for defect detection models.
- Trained machine learning (ML) models and ensembles (XGBoost, LightGBM, random forests, SVR, neural nets) to **R^2 0.9**.
- Delivered ML-driven quality gains projected to cut testing costs by over **$1M/year**; offered a **three-month extension**.

Google — *Open-Source Contributor, Smart Buildings Simulator (SBSim)* Remote | 12/2025 – Present

- Co-leading a benchmark of SBSim against EnergyPlus on **RL training speed** with **Google** and **Berkeley Lab**.
- Proposed the **predictive control** workstream and presented **two talks** to the Google-led working group.
- Contributed code and **reviewed pull requests** for the open-source codebase.

Purdue Energy & Utilities — *Energy Systems Researcher* West Lafayette, IN | 02/2026 – Present

- Designed and ran a **three-day pre-cooling trial** in a 13-zone campus building to shift cooling demand off-peak.
- Delivered a nine-year energy-use assessment of 21 renovations to **Purdue leadership** for the Giant Leaps Master Plan.
- Quantified **~$41k/yr** savings and a statistically significant **8%** lower approach from a two-plant air/dirt-separator retrofit.

UBC — *Graduate Research Assistant, TMMF Lab & FortisBC Renewable Gas* BC, Canada | 05/2021 – 08/2023

- Ran CFD simulations of hydrogen blending into natural gas pipelines; built ideal- and real-gas mixture models in C.
- Delivered **1** journal paper, **3** conference papers, an MS thesis, and **2** awards on a project that seeded the UBC H_2Lab.

SELECTED PUBLICATIONS

Full list on Google Scholar

5. W.G. Dierking, **A.J. Khabbazi**, L.D. Reyes Premer, and K.J. Kircher. Scalable supervisory HVAC control for linear objectives. *65th IEEE Conference on Decision and Control (CDC)*, 2026.
4. L.D. Reyes Premer, **A.J. Khabbazi**, and K.J. Kircher. Indirect data-driven predictive control and the state-space predictor. arXiv:2602.10936, 2026. Under review.
3. **A.J. Khabbazi** and K.J. Kircher. Small HVAC control demonstrations in larger buildings often overestimate savings. *American Control Conference (ACC)*, 2026.
2. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, P. Papageorgiou, A.H. Lee, J.E. Braun, G.P. Henze, and K.J. Kircher. Lessons learned from field demonstrations of model predictive control and reinforcement learning for residential and commercial HVAC: A review. *Applied Energy*, 2025.
1. E.N. Pergantis, L.D. Reyes Premer, **A.J. Khabbazi**, Priyadarshan, F. Wu, D. Ziviani, and K.J. Kircher. Active current limiting control of residential appliances for breaker panel protection across the US: A preliminary study. *CISBAT*, 2025.

SKILLS

AI Methods: Reinforcement Learning (PPO, SAC), Supervised and Unsupervised Learning, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG)
Control & Optimization: Model Predictive Control, Data-Driven Predictive Control, Convex Optimization, System Identification
Frameworks: CVXPY, TensorFlow, TF-Agents, scikit-learn, XGBoost, Stable-Baselines3, PyTorch, Gurobi
Simulation: Google SBSim, Gymnasium, EnergyPlus, BOPTEST, OCHRE, TRNSYS, EES, Ansys Fluent
Programming & Tools: Python (NumPy, SciPy, Pandas), MATLAB, C, SQL, MySQL, InfluxDB, Grafana, Git, Linux, Docker

SELECT HONORS & AWARDS

ASHRAE Graduate Student Grant-in-Aid Award 2025 | Best PhD Forum Presentation Runner-up, ACM BuildSys 2025 | ACM SIGEnergy Student Travel Grant 2025 | Best Paper, CSME Congress 2023 | Best Presentation, CSME Congress 2022